Python Practice: Tuples

Part 2 of 5 | 10 questions | Beginner to Expert

RULES: No for/while loops, no comprehensions, no recursion, no map/filter/reduce, and no hardcoded answers. Use indexing, slicing, operators, built-in functions, and data structure methods.

LEVEL: BEGINNER

Q11. Tuple Indexing

```
colors = ("red", "green", "blue", "yellow", "purple")
```

Extract second, last, and middle three elements.

Q12. Tuple Unpacking

```
employee = ("Ali", 28, "Data Engineer", 150000)
```

Unpack into four variables and display name and salary.

Q13. Tuple Concatenation

```
tuple1 = (10, 20, 30)
tuple2 = (40, 50, 60)
```

Combine tuples, then create another tuple in reverse order.

LEVEL: INTERMEDIATE

Q14. Count and Locate

```
numbers = (10, 20, 30, 20, 40, 20, 50)
```

Count occurrences of 20 and find the first index.

Q15. Extended Tuple Unpacking

```
data = (100, 200, 300, 400, 500, 600)
```

Unpack first two and last into variables; put all remaining elements into a list.

Q16. Modify an Immutable Tuple

```
numbers = (10, 20, 30, 40, 50)
```

Create a new tuple replacing 30 with 300.

Q17. Nested Tuple Access

```
students = (
    ("Ali", (85, 90, 88)),
    ("Sara", (92, 87, 95)),
    ("Ahmed", (78, 84, 80))
)
```

Extract Sara second score, Ali average, and Ahmed highest score.

LEVEL: ADVANCED

Q18. Tuple Sorting

```
employees = (
    ("Ali", 80000),
    ("Sara", 120000),
    ("Ahmed", 95000)
)
```

Sort by salary descending without loops, comprehensions, or lambda.

Q19. Tuple Matrix

```
matrix = ((2, 4, 6), (8, 10, 12), (14, 16, 18))
```

Extract diagonal, calculate its product, and build a tuple of row sums.

Q20. Inventory Tuple Challenge

```
records = (
    ("Laptop", 120000, 5),
    ("Mouse", 1500, 20),
    ("Keyboard", 3500, 10)
)
```

Calculate inventory value, find most expensive product, and build reversed tuple of names.